§2. Basic Descent Methods

Let us turn to the description of some basic techniques used for iteratively solving unconstrained minimization problems. These techniques are of practical importance since they often offer the simplest and more direct alternatives for obtaining solutions.

Underlying Structure of Descent Algorithms

Start at an initial point, determine a direction of movement according to certain principle, then move in this direction to a local minimum of the objective function on that line. At the new point a new direction is determined and the process is repeated.

REMARKS.

- The differences between algorithms depend on the principle by which successive directions of movement are selected.
- Once the selection of a direction is made, all algorithms call for movement to the minimum point on the corresponding line.

The process of determining the minimum point on a given line is called line search. Since line search is actually a procedure for solving one-dimensional minimization problems, it is also called one-dimensional search. In this course, high dimensional minimization problems are frequently solved by executing a sequence of successive line searches.

2.1°. One-Dimensional Search

Consider the line search problem

$$\begin{array}{ll} \text{Min} & f(x) \\ \text{s.t.} & a \le x \le b. \end{array}$$

Since the exact location of the minimum point of f over $[a, b]$ is not known, this interval is called the interval of uncertainty. During the search procedure if we can exclude portions of this interval that do not contain the minimum point, then the interval of uncertainty is reduced. In general, an interval I is called the interval of uncertainty if a minimum point lies in I, though its exact value is unknown.

In practice, when we implement an algorithm, usually we impose a limit $\ell > 0$ on the length of the interval of uncertainty. If the length $\le \ell$, then the algorithm is terminated and any point in this interval can be output as a solution. This ℓ is usually called an allowable final length of uncertainty, which reflects the precision of the algorithm.

EXAMPLE 2.1. Let $f(x)$ be a convex function defined on $[0, 1]$. Suppose the following information is given: $f(0) = 0.25$, $f(0.2) = 0.09$, $f(0.3) = 0.04$, $f(0.6) = 0.01$, $f(1) = 0.25$, and $f'(0.6) = 0.2$. Show that $[0.3, 0.6]$ is an interval of uncertainty.

PROOF. To justify the statement, we need to exclude both $[0, 0.3)$ and $(0.6, 1]$.

For any $x \in [0, 0.3)$, write $0.3 = \alpha \cdot x + (1-\alpha) \cdot 0.6$ for some $0 < \alpha < 1$. In view of the convexity of f, we have $f(0.3) \le \alpha f(x) + (1-\alpha) f(0.6)$, and thus

$$\begin{aligned} f(x) &\ge \frac{1}{\alpha}[f(0.3) - (1-\alpha)f(0.6)] \\ &> \frac{1}{\alpha}[f(0.3) - (1-\alpha)f(0.3)] \qquad (\text{since } f(0.3) = 0.04 \text{ and } f(0.6) = 0.01) \\ &= f(0.3). \end{aligned}$$

Hence $[0, 0.3)$ contains no minimum point. For any $x \in (0.6, 1]$, applying the same proof of Theorem 1.10, we see that $f(x) \ge f(0.6) + f'(0.6)(x - 0.6) > f(0.6)$ as $f'(0.6) > 0$. Hence $(0.6, 1]$ contains no minimum point either. □

REMARK. We made use of the following statement in our solution: Let f be a convex function defined on a convex set Ω. If f is differentiable at a point $\mathbf{x} \in \Omega$, then $f(\mathbf{y}) \ge f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$ for any $\mathbf{y} \in \Omega$. The proof goes along the same line as the only if part of Theorem 1.10.

a) Line Search without Using Derivatives

We begin by defining unimodal functions, which are extremely important in establishing convergence and optimality of the methods presented in this section.

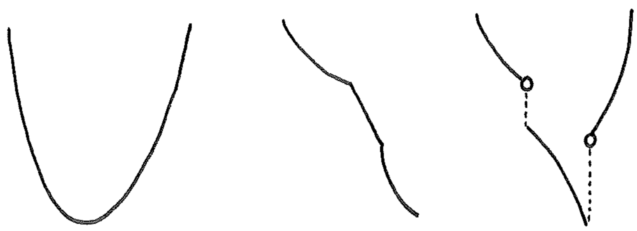
Definition 2.1. *A function f is said to be unimodal on an interval $[a, b]$ if there exists $x^* \in [a, b]$ such that*

(i) *x^* minimizes f on $[a, b]$;*

(ii) *For any x_1, x_2 in $[a, b]$ with $x_1 < x_2$, the following holds:*

- *$x_2 \le x^*$ implies that $f(x_1) > f(x_2)$.*
- *$x^* \le x_1$ implies that $f(x_2) > f(x_1)$.*

REMARK. Unimodal functions are not necessarily continuous or differentiable.



Examples of Unimodal Functions

EXAMPLE 2.2. Let $f(x)$ be a continuous and strictly convex function defined on $[a, b]$. Then f is unimodal.

Proof. Since f is continuous on $[a, b]$, it achieves minimum on some point $x^* \in [a, b]$. Hence (i) of Definition 2.1 follows. Observe that x^* is the unique global minimum point of f on $[a, b]$. Suppose the contrary: $f(x) = f(x^*)$ for some $x \neq x^*$. Let y be a point strictly between x and x^* and write $y = \alpha x + (1-\alpha)x^*$ for some $\alpha \in (0, 1)$. Then by strict convexity $f(y) < \alpha f(x) + (1-\alpha)f(x^*) = f(x^*)$, contradicting the fact that x^* is a minimum point of f.

To justify (ii), let x_1, x_2 be any two points in $[a, b]$ with $x_1 < x_2$, we aim to prove that $x_2 \leq x^*$ implies $f(x_1) > f(x_2)$. According to the above observation, $f(x_1) > f(x^*)$. So we may assume that $x_2 \neq x^*$ and thus $x_2 = \alpha x_1 + (1-\alpha)x^*$ for some $\alpha \in (0, 1)$. Now it follows from the strict convexity that $f(x_2) < \alpha f(x_1) + (1-\alpha)f(x^*) \leq \alpha f(x_1) + (1-\alpha)f(x_1) = f(x_1)$, as desired.

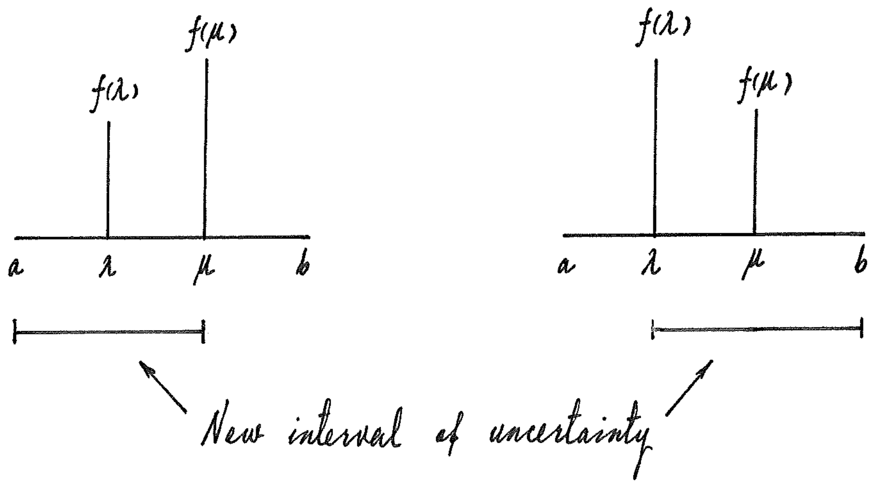
The case $x^* \leq x_1$ can be justified likewise. □

The following facts are crucial to the algorithms given in this section. Keep in mind that f is a unimodal function on $[a, b]$ and x^* is a minimum point of f on $[a, b]$.

FACT 1. x^* is the unique local minimum point of f on $[a, b]$.

Proof. Suppose to the contrary that y^* is a local minimum point of f other than x^*. Without loss of generality, we may assume $x^* < y^*$. By the minimality of y^*, there exists a z such that $x^* < z < y^*$ and $f(z) \geq f(y^*)$. Let $x_1 = z$ and $x_2 = y^*$. Then $x^* < x_1$. By (ii) of Definition 2.1, we have $f(x_2) > f(x_1)$, that is, $f(y^*) > f(z)$, a contradiction. □

FACT 2. Let λ, μ be two points in $[a, b]$ with $\lambda < \mu$. If $f(\lambda) \leq f(\mu)$ then $x^* \in [a, \mu]$; if $f(\lambda) \geq f(\mu)$ then $x^* \in [\lambda, b]$. □



REMARKS.

1. The smallest number of functional evaluations that is needed to reduce the interval

of uncertainty is two.

2. The evaluation of the function on two points acts as an "oracle" (or "fortune-teller") which can guide the direction of new line search.
3. Fact 2 is the core of the succeeding line search methods.

FACT 3. Let λ, μ be two points in $[a, b]$ with $\lambda < \mu$. If $f(\lambda) = f(\mu)$ then $x^* \in [\lambda, \mu]$. □

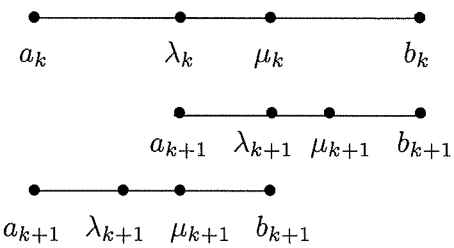
ASSUMPTION. The function involved in each of the following two methods is unimodal.

Method 1. Golden Section Method

Let $[a_k, b_k]$ be the interval of uncertainty at a general iteration k, and let λ_k and μ_k be two points in $[a_k, b_k]$, with $\lambda_k < \mu_k$, at which functional evaluations are to be made. Then the new interval of uncertainty $[a_{k+1}, b_{k+1}] = [\lambda_k, b_k]$ if $f(\lambda_k) > f(\mu_k)$ and $[a_{k+1}, b_{k+1}] = [a_k, \mu_k]$ otherwise. The selection of λ_k and μ_k is under the guidance of the following principles.

1°. The length of the new interval of uncertainty $b_{k+1} - a_{k+1}$ does not depend on the outcome of the k^{th} iteration, that is, on whether $f(\lambda_k) > f(\mu_k)$ or $f(\lambda_k) \leq f(\mu_k)$. In addition, for all iterations k, the reduction ratio $\frac{b_{k+1} - a_{k+1}}{b_k - a_k}$ is always a fixed constant α and

2°. As λ_{k+1} and μ_{k+1} are selected for the purpose of a new iteration, either λ_{k+1} coincides with μ_k or μ_{k+1} coincides with λ_k. (If this can be realized, then during iteration $k+1$, only one extra observation is needed.)



Case 1. $f(\lambda_k) > f(\mu_k)$

Case 2. $f(\lambda_k) \leq f(\mu_k)$

Analysis

According to principle 1°, we have

$$b_k - \lambda_k = \mu_k - a_k = \alpha(b_k - a_k). \tag{2.1}$$

Hence

$$\lambda_k = a_k + (1 - \alpha)(b_k - a_k) \tag{2.2}$$

and

$$\mu_k = a_k + \alpha(b_k - a_k). \tag{2.3}$$

Let us apply principle 2°. In case 1, $a_{k+1} = \lambda_k$ and $b_{k+1} = b_k$. Since $\lambda_{k+1} = \mu_k$, by (2.2) with k replaced by $k+1$, we have

$$\mu_k = \lambda_{k+1} = a_{k+1} + (1-\alpha)(b_{k+1} - a_{k+1}) = \lambda_k + (1-\alpha)(b_k - \lambda_k).$$

Substituting (2.2) and (2.3) into the above equation, we get

$$\alpha^2 + \alpha - 1 = 0 \tag{2.4}$$

In case 2, we obtain the same equation similarly. Since $\alpha \in (0, 1)$, equation (2.4) yields $\alpha = \frac{\sqrt{5}-1}{2} \doteq 0.618$.

Algorithm

Initialization Step Choose an allowable final length of uncertainty $\ell > 0$. Let $[a_1, b_1] = [a, b]$, $\lambda_1 = a_1 + (1-\alpha)(b_1 - a_1)$, and $\mu_1 = a_1 + \alpha(b_1 - a_1)$, where $\alpha = 0.618$, and let $k = 1$, goto the main step.

Main Step

1. If $b_k - a_k \le \ell$, stop; the minimum point lies in $[a_k, b_k]$. Otherwise, if $f(\lambda_k) > f(\mu_k)$, goto step 2; and if $f(\lambda_k) \le f(\mu_k)$, goto step 3;
2. Set $a_{k+1} = \lambda_k$, $b_{k+1} = b_k$, $\lambda_{k+1} = \mu_k$, and $\mu_{k+1} = a_{k+1} + \alpha(b_{k+1} - a_{k+1})$. Evaluate $f(\mu_{k+1})$ and goto step 4;
3. Set $a_{k+1} = a_k$, $b_{k+1} = \mu_k$, $\mu_{k+1} = \lambda_k$, and $\lambda_{k+1} = a_{k+1} + (1-\alpha)(b_{k+1} - a_{k+1})$. Evaluate $f(\lambda_{k+1})$ and goto step 4;
4. Replace k by $k+1$ and goto step 1.

REMARK 1. The reduction ratio $(b_{k+1} - a_{k+1})/(b_k - a_k)$ is always 0.618.

REMARK 2. During each new iteration, only one extra observation is needed. Thus the computational complexity is reduced and efficiency is improved.

REMARK 3. Estimate of the number k of iterations. Since $b_{k+1} - a_{k+1} = \alpha(b_k - a_k) = \cdots = \alpha^k(b_1 - a_1)$, we have $(0.618)^k \le \frac{\ell}{b_1 - a_1}$.

REMARK 4. Estimate of the number n of evaluations:

$$(0.618)^{n-1} \le \frac{\ell}{b_1 - a_1}. \tag{2.5}$$

EXAMPLE 2.4. Solve the following problem by the golden section method; the allowable final length of interval is 0.2.

$$\begin{array}{ll} \text{Min} & x^2 + 2x \\ \text{s.t.} & -3 \le x \le 5. \end{array}$$

SOLUTION. Clearly the function f to be minimized is unimodal (why ?), and the initial interval of uncertainty is of length 8. We need to reduce this interval to one whose length is at most 0.2. The first two observations are located at $\lambda_1 = -3 + 0.382 \times 8 = 0.056$, $\mu_1 = -3 + 0.618 \times 8 = 1.944$. Note that $f(\lambda_1) < f(\mu_1)$. Hence the new interval of uncertainty is $[-3, 1.944]$. The process is repeated, and the computations are summarized in the following table

Iteration k	a_k	b_k	λ_k	μ_k	$f(\lambda_k)$	$f(\mu_k)$
1	−3.000	5.000	0.056	1.944	0.115*	7.667*
2	−3.000	1.944	−1.112	0.056	−0.987*	0.115
3	−3.000	0.056	−1.832	−1.112	−0.308*	−0.987
4	−1.832	0.056	−1.112	−0.664	−0.987	−0.887*
5	−1.832	−0.664	−1.384	−1.112	−0.853*	−0.987
6	−1.384	−0.664	−1.112	−0.936	−0.987	−0.996*
7	−1.112	−0.664	−0.936	−0.840	−0.996	−0.974*
8	−1.112	−0.840	−1.016	−0.936	−1.000*	−0.996
9	−1.112	−0.936				

where the values of f that are computed at each iteration are indicated by an asterisk. After eight iterations involving nine observations, the interval of uncertainty is $[-1.112, -0.936]$, so the minimum can be estimated to be any point in it. Note that the true minimum is in fact -1.0. □

Method 2. Fibonacci Search Method

Similar to the golden section method, the Fibonacci search makes two functional evaluations at the first iteration and then only one evaluation at each of the subsequent iterations. However, the reduction ratio of the Fibonacci search method varies from one iteration to another.

The method is based on the Fibonacci sequence $\{F_n\}$ defined as follows.

$$F_{n+1} = F_n + F_{n-1}, \ n = 1, 2, \cdots$$
$$F_0 = F_1 = 1.$$

The sequence is therefore $1, 1, 2, 3, 5, \cdots$

Proposition 2.1. $F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}\right]$ *for* $n = 0, 1, 2, \cdots$.

Proposition 2.2. $F_n > \left(\frac{1+\sqrt{5}}{2}\right)^{n-1}$ *for* $n = 2, 3, \cdots$.

REMARK. Unlike the Golden Section method, the Fibonacci search method requires the total number n of observations be chosen beforehand.

Suppose the interval of uncertainty is $[a_k, b_k]$ at iteration k. Define

$$\lambda_k = a_k + \frac{F_{n-k-1}}{F_{n-k+1}}(b_k - a_k) \tag{2.6}$$

$$\mu_k = a_k + \frac{F_{n-k}}{F_{n-k+1}}(b_k - a_k) \tag{2.7}$$

for $k = 1, 2, \cdots, n-1$. The new interval of uncertainty $[a_{k+1}, b_{k+1}] = [\lambda_k, b_k]$ if $f(\lambda_k) > f(\mu_k)$ and $[a_{k+1}, b_{k+1}] = [a_k, \mu_k]$ otherwise. Then

$$b_{k+1} - a_{k+1} = \frac{F_{n-k}}{F_{n-k+1}}(b_k - a_k). \tag{2.8}$$

Indeed, if $[a_{k+1}, b_{k+1}] = [\lambda_k, b_k]$, then we have

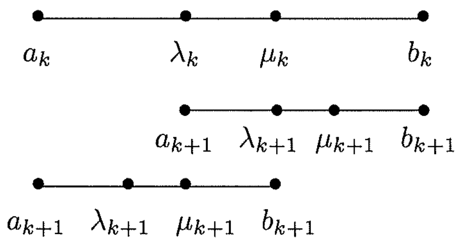
$$\begin{aligned} b_{k+1} - a_{k+1} &= b_k - a_k - \frac{F_{n-k-1}}{F_{n-k+1}}(b_k - a_k) \\ &= \frac{F_{n-k+1} - F_{n-k-1}}{F_{n-k+1}}(b_k - a_k) \\ &= \frac{F_{n-k}}{F_{n-k+1}}(b_k - a_k). \end{aligned}$$

Similarly, we can check the other case.

Now let us show that if $[a_{k+1}, b_{k+1}] = [\lambda_k, b_k]$ then $\lambda_{k+1} = \mu_k$; if $[a_{k+1}, b_{k+1}] = [a_k, \mu_k]$ then $\mu_{k+1} = \lambda_k$.

Case 1.

Case 2.



Indeed, if $[a_{k+1}, b_{k+1}] = [\lambda_k, b_k]$, then

$$\begin{aligned} \lambda_{k+1} &= a_{k+1} + \tfrac{F_{n-k-2}}{F_{n-k}}(b_{k+1} - a_{k+1}) \\ &= \lambda_k + \tfrac{F_{n-k-2}}{F_{n-k}}(b_k - \lambda_k) \end{aligned}$$

Using (2.6), we have

$$\begin{aligned} \lambda_{k+1} &= a_k + \frac{F_{n-k-1}}{F_{n-k+1}}(b_k - a_k) + \frac{F_{n-k-2}}{F_{n-k}}\left(1 - \frac{F_{n-k-1}}{F_{n-k+1}}\right)(b_k - a_k) \\ &= a_k + \frac{F_{n-k-1} + F_{n-k-2}}{F_{n-k+1}}(b_k - a_k) \\ &= a_k + \frac{F_{n-k}}{F_{n-k+1}}(b_k - a_k) \\ &= \mu_k. \end{aligned}$$

Similarly we can verify the other case. Thus, in either case, only one observation is needed at iteration $k+1$.

Analysis

At the first iteration, two observations are made, and at each subsequent iteration, only one observation is necessary. Thus at the end of iteration $n-2$, we have completed $n-1$ functional evaluations. For $k = n-1$, it follows from (2.6) and (2.7) that $\lambda_{n-1} = \mu_{n-1} = \frac{1}{2}(a_{n-1} + b_{n-1})$. Since either $\lambda_{n-1} = \mu_{n-2}$ or $\mu_{n-1} = \lambda_{n-2}$, theoretically no new observations are to be made at this stage. However, in order to further reduce the interval of uncertainty, the last observation is placed slightly to the right or to the left of the midpoint $\lambda_{n-1} = \mu_{n-1}$, so that $\frac{1}{2}(b_{n-1} - a_{n-1})$ is the length of the final interval of uncertainty $[a_n, b_n]$.

REMARK. Estimate of the number n of evaluations:

$$\frac{1}{F_n} \leq \frac{l}{b_1 - a_1}. \tag{2.9}$$

This inequality follows instantly from (2.8) as $b_n - a_n = \frac{1}{F_n}(b_1 - a_1)$.

Algorithm

Initialization Step Choose an allowable final length of uncertainty $\ell > 0$ and a distinguishability constant $\varepsilon > 0$. Let $[a_1, b_1] = [a, b]$ and choose the number n of observations such that $F_n \geq \frac{b_1 - a_1}{\ell}$. Let $\lambda_1 = a_1 + \frac{F_{n-2}}{F_n}(b_1 - a_1)$ and $\mu_1 = a_1 + \frac{F_{n-1}}{F_n}(b_1 - a_1)$, and let $k = 1$, goto the main step.

Main Step

1. If $f(\lambda_k) > f(\mu_k)$, goto step 2; otherwise, goto step 3.
2. Set $a_{k+1} = \lambda_k$, $b_{k+1} = b_k$, $\lambda_{k+1} = \mu_k$, and $\mu_{k+1} = a_{k+1} + \frac{F_{n-k-1}}{F_{n-k}}(b_{k+1} - a_{k+1})$. If $k = n-2$, goto Step 5; otherwise, goto Step 4.
3. Set $a_{k+1} = a_k$, $b_{k+1} = \mu_k$, $\lambda_{k+1} = a_{k+1} + \frac{F_{n-k-2}}{F_{n-k}}(b_{k+1} - a_{k+1})$, and $\mu_{k+1} = \lambda_k$. If $k = n-2$, goto Step 5; otherwise, goto Step 4.
4. Replace k by $k+1$ and goto Step 1.
5. Let $\lambda_n = \lambda_{n-1}$ and $\mu_n = \lambda_{n-1} + \varepsilon$. Set $a_n = \lambda_n$ and $b_n = b_{n-1}$ if $f(\lambda_n) > f(\mu_n)$ and set $a_n = a_{n-1}$ and $b_n = \lambda_n$ otherwise, stop; the minimum point is in $[a_n, b_n]$.

EXAMPLE 2.5. Solve the following problem by the Fibonacci Method; the allowable final length of interval is 0.2.

$$\begin{array}{ll} \text{Min} & x^2 + 2x \\ \text{s.t.} & -3 \le x \le 5 \end{array}$$

SOLUTION. We aim to reduce the interval of uncertainty to one whose length is at most 0.2, so $F_n \ge 8/0.2 = 40$, and hence $n = 9$. We adopt the distinguishability number $\varepsilon = 0.01$.

The first two observations are located at $\lambda_1 = -3 + \frac{F_7}{F_9} \times 8 = 0.054545$, $\mu_1 = -3 + \frac{F_8}{F_9} \times 8 = 1.945454$. Note that $f(\lambda_1) < f(\mu_1)$. Hence the new interval of uncertainty is $[-3, 1.945454]$. The process is repeated, and the computations are summarized in the following table

Iteration k	a_k	b_k	λ_k	μ_k	$f(\lambda_k)$	$f(\mu_k)$
1	−3.000000	5.000000	0.054545	1.945454	0.112065*	7.675699*
2	−3.000000	1.945454	−1.109091	0.054545	−0.988099*	0.112065
3	−3.000000	0.054545	−1.836363	−1.109091	−0.300497*	−0.988099
4	−1.836363	0.054545	−1.109091	−0.672727	−0.988099	−0.892892*
5	−1.836363	−0.672727	−1.399999	−1.109091	−0.840001*	−0.988099
6	−1.399999	−0.672727	−1.109091	−0.963636	−0.988099	−0.998677*
7	−1.109091	−0.672727	−0.963636	−0.818182	−0.998677	−0.966942*
8	−1.109091	−0.818182	−0.963636	−0.963636	−0.998677	−0.998677
9	−1.109091	−0.963636	−0.963636	−0.953636	−0.998677	−0.997850*

where the values of f that are computed at each iteration are indicated by an asterisk. Note that $\lambda_8 = \mu_8 = \lambda_7 = -0.963636$. So $\lambda_9 = \lambda_8$ and $\mu_9 = \lambda_8 + \varepsilon = -0.953636$. Since $f(\mu_9) > f(\lambda_8)$, the final interval of uncertainty $[a_9, b_9]$ is $[-1.109091, -0.963636]$. □

Comparison of Line Search Methods without Using Derivatives

Let n stand for the required number of observations. Then n measures the computational complexity of a line search method. In view of (2.5) and (2.9), we have

- Golden Section: $(0.618)^{n-1} \le \frac{\ell}{b-a}$.
- Fibonacci Search: $\frac{1}{F_n} \le \frac{\ell}{b-a}$.

Note that, by Proposition 2.2, $\frac{1}{F_n} < (0.618)^{n-1}$. For a fixed ratio $\frac{b-a}{\ell}$, the smaller the number of observations required, the more efficient the algorithm is. It is evident that, theoretically, the Fibonacci method is more efficient than the golden section method. However, practically, the golden section method is more widely used.

b) Line Search Using Derivatives

The search methods covered in the last section are all based on the assumption that the function being searched is unimodal. In most problems, however, it can be safely assumed that the function being searched possess a certain degree of smoothness, and one might therefore expect that more efficient search techniques exploiting this smoothness can be devised; and indeed they can.

Method 1. Newton's Method

Let f be a single variable function to be minimized. Suppose that at a point x_k, where a measurement is made, it is possible to evaluate the three numbers $f(x_k)$, $f'(x_k)$, and $f''(x_k)$. Newton's method is based on exploiting the quadratic approximations of the function f at a given point x_k, that is

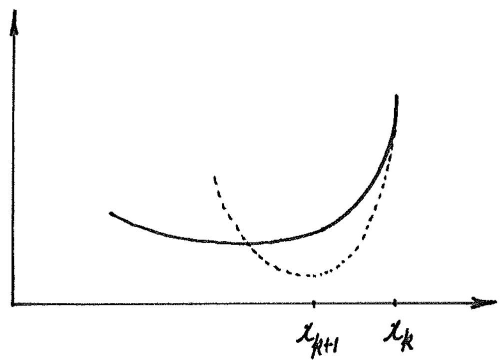
$$q(x) = f(x_k) + f'(x_k)(x - x_k) + \frac{1}{2} f''(x_k)(x - x_k)^2.$$

The point x_{k+1} is taken to be the point where $q'(x) = 0$, or $f'(x_k) + f''(x_k)(x_{k+1} - x_k) = 0$. Hence

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}. \tag{2.10}$$

The procedure is terminated when $|x_{k+1} - x_k| < \varepsilon$ or when $|f'(x_{k+1})| < \varepsilon$, where ε is a prespecified termination scalar.

The process is illustrated in the following figure.



EXAMPLE 2.6. Solve the problem via Newton's method

$$\text{Minimize} \quad f(x) = \begin{cases} 4x^3 - 3x^4, & \text{if } x \geq 0 \\ 4x^3 + 3x^4, & \text{otherwise.} \end{cases}$$

The termination scalar is 0.001 and the starting point is 0.4.

SOLUTION. Note that

$$f'(x) = \begin{cases} 12x^2 - 12x^3, & \text{if } x \geq 0 \\ 12x^2 + 12x^3, & \text{else} \end{cases}$$

$$f''(x) = \begin{cases} 24x - 36x^2, & \text{if } x \geq 0 \\ 24x + 36x^2, & \text{else.} \end{cases}$$

Hence $f \in C^2$. Let us apply Newton's method, starting from $x_0 = 0.4$. The computation is summarized in the following table.

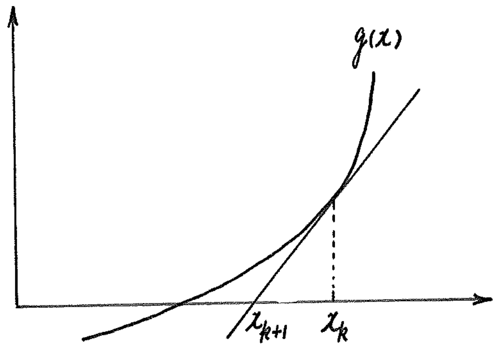
Iteration k	x_k	$f'(x_k)$	$f''(x_k)$	x_{k+1}
0	0.400000	1.152000	3.840000	0.100000
1	0.100000	0.108000	2.040000	0.047059
2	0.047059	0.025324	1.049692	0.022934
3	0.022934	0.006167	0.531481	0.011331
4	0.011331	0.001523	0.267322	0.005634
5	0.005634	0.000379		

The procedure produces the point 0.005634 after five iterations. (It can be shown that the procedure indeed converges to the stationary point $x = 0$.) $\square$

Since x_{k+1} resulting from (2.10) does not depend on $f(x_k)$, this method can be viewed as a technique for iteratively solving equations of the form $g(x) = 0$, where, when applied to the minimization problem, we put $g(x) \equiv f'(x)$. In this notation, Newton's method takes the form

$$x_{k+1} = x_k - \frac{g(x_k)}{g'(x_k)}, \tag{2.11}$$

which is illustrated in the following figure.



Theorem 2.1. *Let $g \in C^2$ and let x^* satisfy $g(x^*) = 0$, $g'(x^*) \neq 0$. Then provided x_0 is sufficiently close to x^*, the sequence $\{x_k\}$ generated by Newton's method* (2.11) *converges to x^* with an order of convergence at least two.*

Proof. By hypothesis, there exist two positive numbers c_1 and c_2 such that $|g''(x)| < c_1$ and $|g'(x)| > c_2$ for all x sufficiently close to x^*. Since $g(x^*) = 0$, we can write

$$\begin{aligned} x_{k+1} - x^* &= x_k - x^* - \frac{g(x_k) - g(x^*)}{g'(x_k)} \\ &= [g(x^*) - g(x_k) - g'(x_k)(x^* - x_k)]/g'(x_k). \end{aligned}$$

By Taylor's theorem, $g(x^*) - g(x_k) - g'(x_k)(x^* - x_k) = \frac{1}{2}g''(\xi)(x^* - x_k)^2$ for some ξ between x^* and x_k. Thus in the region near x^*, we have

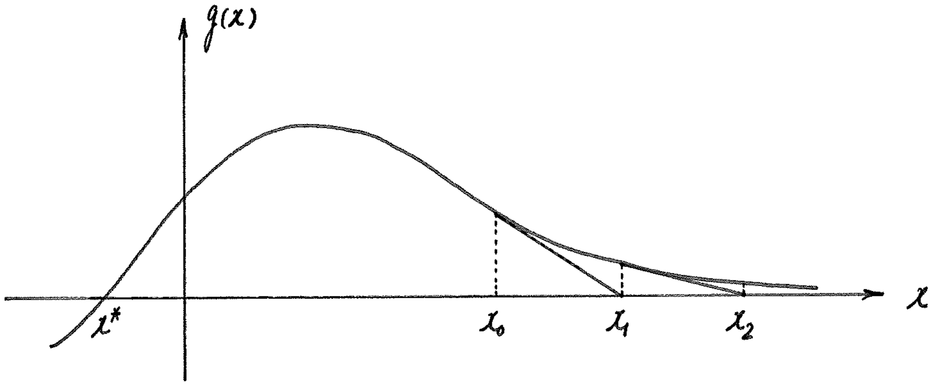
$$\begin{aligned} |x_{k+1} - x^*| &= \frac{1}{2}|g''(\xi)|(x_k - x^*)^2/|g'(x_k)| \\ &\leq \frac{c_1}{2c_2}|x_k - x^*|^2. \end{aligned} \tag{2.12}$$

Suppose x_0 is chosen so that $\frac{c_1}{2c_2}|x_0 - x^*| \triangleq c < 1$. From (2.12), we see that $|x_k - x^*| \leq c^k|x_0 - x^*|$, so x^* is the limit of $\{x_k\}$. Since

$$\overline{\lim_{k\to\infty}} \frac{|x_{k+1} - x^*|}{|x_k - x^*|^2} \leq \frac{c_1}{2c_2},$$

Newton's method converges to x^* with an order at least two. □

REMARK. Newton's method is not globally convergent.



2.2° <u>Multidimensional Search</u>

Let us now discuss two methods that apply derivatives in determining the search directions. They are: the <u>steepest descent method</u> and <u>Newton's Method</u>.

<u>Method 1.</u> The Steepest Descent Method

This method is one of the most fundamental procedures for minimizing a differentiable multivariate function. More advanced algorithms are often motivated by an attempt to modify the basic steepest descent techniques in such a way that the new algorithm will have superior convergence properties.

Definition 2.2. *A vector* $\mathbf{d}$ *is called a* <u>*direction of descent*</u> *of a function* f *at* $\mathbf{x}$ *if there exists* $\delta > 0$ *such that* $f(\mathbf{x} + \alpha\mathbf{d}) < f(\mathbf{x})$ *for all* $0 < \alpha < \delta$.

Clearly, if

$$f'(\mathbf{x},\mathbf{d}) \triangleq \lim_{\alpha\to 0^+} \frac{f(\mathbf{x}+\alpha\mathbf{d}) - f(\mathbf{x})}{\alpha} < 0,$$

then $\mathbf{d}$ is a direction of descent.

The steepest descent method is a method that moves along the direction $\mathbf{d}$, with $\|\mathbf{d}\| = 1$, that minimizes $f'(\mathbf{x},\mathbf{d})$.

Theorem 2.2. *Let f be differentiable at $\mathbf{x}$ with $\nabla f(\mathbf{x}) \neq 0$. Then the optimal solution to the problem*

$$\begin{array}{ll} \mathit{Min} & f'(\mathbf{x},\mathbf{d}) \\ \mathit{s.t.} & \|\mathbf{d}\| \leq 1 \end{array} \tag{2.13}$$

is given by $\mathbf{d} = -\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$; that is, $-\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$ is the direction of steepest descent of f at $\mathbf{x}$.

Proof. From differentiability of f at $\mathbf{x}$, it follows that

$$f'(\mathbf{x},\mathbf{d}) = \lim_{\alpha\to 0^+} \frac{f(\mathbf{x}+\alpha\mathbf{d}) - f(\mathbf{x})}{\alpha} = \nabla f(\mathbf{x})^T\mathbf{d}.$$

By the Schwartz inequality, for $\|\mathbf{d}\| \leq 1$, we have

$$\nabla f(\mathbf{x})^T\mathbf{d} \geq -\|\nabla f(\mathbf{x})^T\| \cdot \|\mathbf{d}\| \geq -\|\nabla f(\mathbf{x})\|$$

with equality holding throughout iff $\mathbf{d} = -\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$. Thus, $-\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$ is the optimal solution to (2.13). □

The basic idea of the present method is as follows: Given a point $\mathbf{x}$, the steepest descent algorithm proceeds by performing line search along the direction $-\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$ or, equivalently, along the direction $-\nabla f(\mathbf{x})$.

Algorithm

Initialization Step Let $\varepsilon > 0$ be a termination scalar. Choose a starting point $\mathbf{x}_0$. Let $k = 0$, goto the main step.

Main Step If $\|\nabla f(\mathbf{x}_k)\| < \varepsilon$, stop; otherwise, let $\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$, and let α_k be an optimal solution to the problem

$$\begin{array}{ll} \text{Min} & f(\mathbf{x}_k + \alpha\mathbf{d}_k) \\ \text{s.t.} & \alpha \geq 0. \end{array} \tag{2.14}$$

Let $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k\mathbf{d}_k$; replace k by $k+1$, and repeat the main step.

EXAMPLE 2.7. Solve the following problem by the steepest descent method

$$\text{Minimize} \quad f(\mathbf{x}) = (x_1 - 2)^4 + (x_1 - 2x_2)^2.$$

The termination scalar is 0.1 and the starting point is $(0,3)^T$.

SOLUTION. The summary of the computations is given in the following table. Note that $f(\mathbf{x}_k + \alpha\mathbf{d}_k)$ is strictly convex provided $\mathbf{d}_k \neq 0$. So we can find α_k using the golden section method. After seven iterations, the point $\mathbf{x}_7 = (2.28, 1.15)^T$ is reached. The algorithm is terminated since $\|\nabla f(\mathbf{x}_7)\| = 0.09 < 0.1$. In fact that the minimum point of this problem is $(2, 1)^T$.

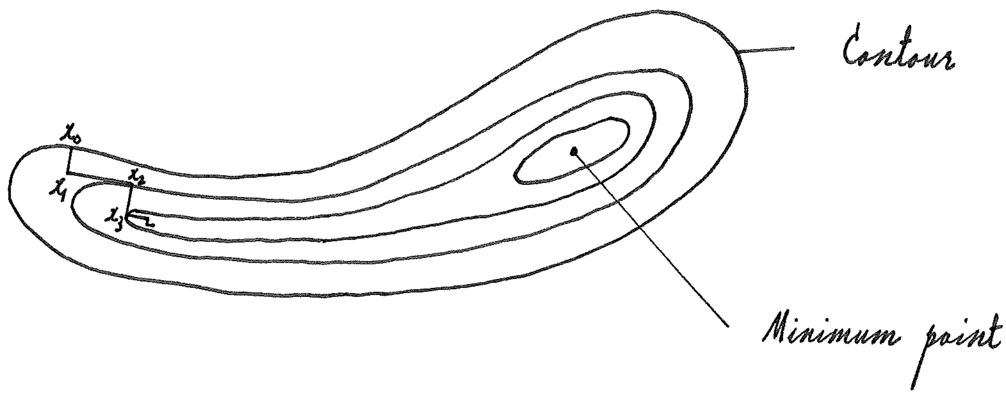
Iteration k	$\mathbf{x}_k$ $f(\mathbf{x}_k)$	$\nabla f(\mathbf{x}_k)$	$\|\nabla f(\mathbf{x}_k)\|$	$\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$	α_k	$\mathbf{x}_{k+1}$
0	(0.00, 3.00) 52.00	(−44.00, 24.00)	50.12	(44.00, −24.00)	0.062	(2.70, 1.51)
1	(2.70, 1.51) 0.34	(0.73, 1.28)	1.47	(−0.73, −1.28)	0.24	(2.52, 1.20)
2	(2.52, 1.20) 0.09	(0.80, −0.48)	0.93	(−0.80, 0.48)	0.11	(2.43, 1.25)
3	(2.43, 1.25) 0.04	(0.18, 0.28)	0.33	(−0.18, −0.28)	0.31	(2.37, 1.16)
4	(2.37, 1.16) 0.02	(0.30, −0.20)	0.36	(−0.30, 0.20)	0.12	(2.33, 1.18)
5	(2.33, 1.18) 0.01	(0.08, 0.12)	0.14	(−0.08, −0.12)	0.36	(2.30, 1.14)
6	(2.30, 1.14) 0.009	(0.15, −0.08)	0.17	(−0.15, 0.08)	0.13	(2.28, 1.15)
7	(2.28, 1.15) 0.007	(0.05, 0.08)	0.09			

REMARK 1. $\alpha_k > 0$. To justify this, let $g(\alpha) = f(\mathbf{x}_k + \alpha\mathbf{d}_k)$. Then $g'(0) = \mathbf{d}_k^T \nabla f(\mathbf{x}_k) = -\nabla f(\mathbf{x}_k)^T \nabla f(\mathbf{x}_k) < 0$. Hence the minimum value of $g(\alpha)$ occurs at some $\alpha_k > 0$.

REMARK 2. Two consecutive search directions are orthogonal. To see this, let $g(\alpha) = f(\mathbf{x}_k + \alpha\mathbf{d}_k)$. Then $g'(\alpha_k) = 0$ or $\mathbf{d}_k^T \nabla f(\mathbf{x}_k + \alpha_k\mathbf{d}_k) = 0$. Thus $\mathbf{d}_k^T \mathbf{d}_{k+1} = 0$.

REMARK 3. The steepest descent method usually works quite well during early stages of the optimization process, depending on the starting point. However, as a stationary point is approached, the method usually behaves poorly, taking small steps.

The number of necessary steps to minimize ill-conditioned functions of the "narrow and elongated valley" type may be considerable. This phenomenon is called "zigzagging".



The Quadratic Case

Essentially all of the important local convergence characteristics of the steepest descent method are revealed by an investigation of the method when applied to quadratic problems, and the convergence property derived for the quadratic case can be translated into a similar one for nonquadratic case.

Consider the quadratic problem

$$\text{Minimize} \quad f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T Q \mathbf{x} - \mathbf{b}^T \mathbf{x},$$

where Q is a positive definite $n \times n$ matrix.

Lemma 2.1. *The steepest descent algorithm for the quadratic problem is as follows.*

$$\begin{aligned} \mathbf{d}_k &= \mathbf{b} - Q\mathbf{x}_k, \\ \alpha_k &= \frac{\mathbf{d}_k^T \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k}, \\ \mathbf{x}_{k+1} &= \mathbf{x}_k + \frac{\mathbf{d}_k^T \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k}\mathbf{d}_k. \end{aligned} \tag{2.15}$$

Proof. Since $\nabla f(\mathbf{x}) = Q\mathbf{x} - \mathbf{b}$, we have $\mathbf{d}_k = -\nabla f(\mathbf{x}_k) = \mathbf{b} - Q\mathbf{x}_k$, as desired. Next, set $g(\alpha) = f(\mathbf{x}_k + \alpha \mathbf{d}_k)$. Then $g(\alpha)$ is a convex function as f is convex. So any point satisfying $g'(\alpha) = 0$ is a minimum point of $g(\alpha)$; this equation yields

$$\mathbf{x}_k^T Q \mathbf{d}_k + \alpha \mathbf{d}_k^T Q \mathbf{d}_k - \mathbf{d}_k^T \mathbf{b} = 0.$$

From $\mathbf{b} = Q\mathbf{x}_k + \mathbf{d}_k$, it can be seen that

$$\alpha_k = \frac{\mathbf{d}_k^T \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k}.$$

Hence

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \left(\frac{\mathbf{d}_k^T \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k}\right)\mathbf{d}_k.$$

The unique minimum point of f can be found directly, by setting the gradient to zero, as the vector $\mathbf{x}^*$ satisfying $Q\mathbf{x}^* = \mathbf{b}$. Define

$$E(\mathbf{x}) = \frac{1}{2}(\mathbf{x}-\mathbf{x}^*)^T Q(\mathbf{x}-\mathbf{x}^*).$$

Then it is easy to see that $E(\mathbf{x}) = f(\mathbf{x}) + \frac{1}{2}\mathbf{x}^{*^T}Q\mathbf{x}^*$, which shows that the function E differs from f only by a constant. For many purposes then, it will be convenient to consider that we are minimizing E rather than f.

Lemma 2.2. *The iterative process (2.15) satisfies*

$$E(\mathbf{x}_{k+1}) = \left\{1 - \frac{(\mathbf{d}_k^T\mathbf{d}_k)^2}{(\mathbf{d}_k^TQ\mathbf{d}_k)(\mathbf{d}_k^TQ^{-1}\mathbf{d}_k)}\right\}E(\mathbf{x}_k).$$

Proof. Let $\mathbf{y}_k = \mathbf{x}_k - \mathbf{x}^*$. Then $Q\mathbf{y}_k = Q\mathbf{x}_k - Q\mathbf{x}^* = Q\mathbf{x}_k - \mathbf{b} = -\mathbf{d}_k$. Thus we have

$$\begin{aligned}\frac{E(\mathbf{x}_{k+1}) - E(\mathbf{x}_k)}{E(\mathbf{x}_k)} &= \frac{\frac{1}{2}(\mathbf{y}_k+\alpha_k\mathbf{d}_k)^TQ(\mathbf{y}_k+\alpha_k\mathbf{d}_k) - \frac{1}{2}\mathbf{y}_k^TQ\mathbf{y}_k}{\frac{1}{2}\mathbf{y}_k^TQ\mathbf{y}_k}\\
&= \frac{2\alpha_k\mathbf{d}_k^TQ\mathbf{y}_k + \alpha_k^2\mathbf{d}_k^TQ\mathbf{d}_k}{\mathbf{y}_k^TQ\mathbf{y}_k}\\
&= \frac{-2\frac{(\mathbf{d}_k^T\mathbf{d}_k)^2}{\mathbf{d}_k^TQ\mathbf{d}_k} + \frac{(\mathbf{d}_k^T\mathbf{d}_k)^2}{\mathbf{d}_k^TQ\mathbf{d}_k}}{\mathbf{d}_k^TQ^{-1}\mathbf{d}_k}\\
&= -\frac{(\mathbf{d}_k^T\mathbf{d}_k)^2}{(\mathbf{d}_k^TQ\mathbf{d}_k)(\mathbf{d}_k^TQ^{-1}\mathbf{d}_k)}.\end{aligned}$$

It follows that

$$E(\mathbf{x}_{k+1}) = \left\{1 - \frac{(\mathbf{d}_k^T\mathbf{d}_k)^2}{(\mathbf{d}_k^TQ\mathbf{d}_k)(\mathbf{d}_k^TQ^{-1}\mathbf{d}_k)}\right\}E(\mathbf{x}_k).$$

□

Lemma 2.3. *(Kantorovich Inequality)*
Let λ_1 and λ_n be the smallest and largest eigenvalues of Q, respectively. Then for all $\mathbf{x} \neq 0$, there holds

$$\frac{(\mathbf{x}^T\mathbf{x})^2}{(\mathbf{x}^TQ\mathbf{x})(\mathbf{x}^TQ^{-1}\mathbf{x})} \geq \frac{4\lambda_1\lambda_n}{(\lambda_1+\lambda_n)^2}. \tag{2.16}$$

Proof. Since Q is a positive definite matrix, there exists an orthogonal matrix P such that

$$Q = P^T \wedge P, \quad \text{where} \quad \wedge = \begin{pmatrix} \lambda_1 & & & 0\\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{pmatrix}.$$

and $0 < \lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$ are n eigenvalues of Q. Let $\mathbf{y} = P\mathbf{x}$. Then $\mathbf{y}^T\mathbf{y} = \mathbf{x}^T P^T P\mathbf{x} = \mathbf{x}^T\mathbf{x}$. Thus

$$\frac{(\mathbf{x}^T\mathbf{x})^2}{(\mathbf{x}^T Q\mathbf{x})(\mathbf{x}^T Q^{-1}\mathbf{x})} = \frac{(\mathbf{y}^T\mathbf{y})^2}{(\mathbf{y}^T \wedge \mathbf{y})(\mathbf{y}^T \wedge^{-1} \mathbf{y})}.$$

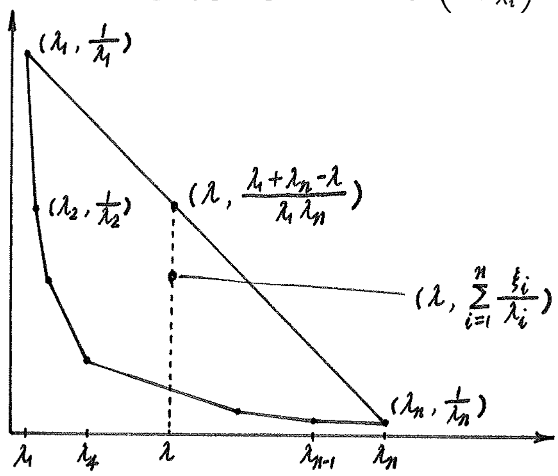
We aim to prove that

$$\frac{(\sum_{i=1}^{n} \lambda_i y_i^2)(\sum_{i=1}^{n} \lambda_i^{-1} y_i^2)}{(\sum_{i=1}^{n} y_i^2)^2} \le \frac{(\lambda_1 + \lambda_n)^2}{4\lambda_1\lambda_n}. \tag{2.17}$$

For this purpose, let $\xi_i = y_i^2 / \sum_{j=1}^{n} y_j^2$. Then $\xi_i \ge 0$ and $\sum_{i=1}^{n} \xi_i = 1$. So (2.17) becomes

$$\left(\sum_{i=1}^{n} \xi_i\lambda_i\right) \cdot \left(\sum_{i=1}^{n} \frac{\xi_i}{\lambda_i}\right) \le \frac{(\lambda_1 + \lambda_n)^2}{4\lambda_1\lambda_n}. \tag{2.18}$$

To justify (2.18), consider the convex polygon generated by $\left(\lambda_i, \frac{1}{\lambda_i}\right)$ for $i = 1, 2 \cdots, n$ as shown below



Let $\lambda = \sum_{i=1}^{n} \xi_i\lambda_i$. Then it can be seen from the graph that

$$\sum_{i=1}^{n} \frac{\xi_i}{\lambda_i} \le \frac{\lambda_1 + \lambda_n - \lambda}{\lambda_1\lambda_n}.$$

Hence

$$\begin{aligned}\left(\sum_{i=1}^{n} \xi_i\lambda_i\right)\left(\sum_{i=1}^{n} \frac{\xi_i}{\lambda_i}\right) &\le \lambda \cdot \frac{\lambda_1 + \lambda_n - \lambda}{\lambda_1\lambda_n} \\ &\le \operatorname*{Max}_{\lambda_1 \le \lambda \le \lambda_n} \lambda \cdot \frac{\lambda_1 + \lambda_n - \lambda}{\lambda_1\lambda_n} \\ &= \frac{(\lambda_1 + \lambda_n)^2}{4\lambda_1\lambda_n},\end{aligned}$$

completing the proof. □

Theorem 2.3. *(Steepest Descent-Quadratic Case)*

For any $\mathbf{x}_0 \in \mathbb{R}^n$*, the steepest descent method converges to the unique minimum point* $\mathbf{x}^*$ *of* f*. Furthermore, with* $E(\mathbf{x}) = \frac{1}{2}(\mathbf{x}-\mathbf{x}^*)^T Q(\mathbf{x}-\mathbf{x}^*)$*, at each step* k *we have*

$$E(\mathbf{x}_{k+1}) \leq \left(\frac{\lambda_n - \lambda_1}{\lambda_n + \lambda_1}\right)^2 E(\mathbf{x}_k). \tag{2.19}$$

Proof. By Lemma 2.2 and the Kantorovich inequality, we have

$$E(\mathbf{x}_{k+1}) \leq \left\{1 - \frac{4\lambda_1\lambda_n}{(\lambda_1+\lambda_n)^2}\right\} E(\mathbf{x}_k) = \left(\frac{\lambda_n - \lambda_1}{\lambda_n + \lambda_1}\right)^2 E(\mathbf{x}_k)$$

It follows that $E(\mathbf{x}_k) \to 0$ and hence $\mathbf{x}_k \to \mathbf{x}^*$ as Q is positive definite. □

REMARK. Formula (2.19) serves as a fundamental reference point for many other algorithms.

Method 2. Newton's Method

The idea behind Newton's method is that the function f being minimized is approximated locally by a quadratic function, and this approximate function is minimized exactly.

At point $\mathbf{x}_k$, let

$$q(\mathbf{x}) = f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T(\mathbf{x}-\mathbf{x}_k) + \frac{1}{2}(\mathbf{x}-\mathbf{x}_k)^T \nabla^2 f(\mathbf{x}_k)(\mathbf{x}-\mathbf{x}_k).$$

Setting $\nabla q(\mathbf{x}) = 0$ gives

$$\mathbf{x}_{k+1} = \mathbf{x}_k - [\nabla^2 f(\mathbf{x}_k)]^{-1}\nabla f(\mathbf{x}_k). \tag{2.20}$$

This equation is the pure form of Newton's Method. The iterative process is terminated when $\|\mathbf{x}_{k+1} - \mathbf{x}_k\| < \varepsilon$ or when $\|\nabla f(\mathbf{x}_{k+1})\| < \varepsilon$, where ε is a prespecified termination scalar. In view of the second-order sufficiency conditions for minimum points, we assume that $\nabla^2 f(\mathbf{x}^*)$ is positive definite at a local minimum point $\mathbf{x}^*$. We can then argue that if $f \in \mathcal{C}^2, \nabla^2 f(\mathbf{x})$ is positive definite near $\mathbf{x}^*$ and hence the method is well defined near the solution.

Theorem 2.4. *(Newton's Method)*

Let $f \in \mathcal{C}^3$ *on* $\mathbb{R}^n$*. Suppose that at a local minimum point* $\mathbf{x}^*$*, the Hessian* $\nabla^2 f(\mathbf{x}^*)$ *is positive definite. Then the sequence generated by Newton's method (2.20) converges to* $\mathbf{x}^*$ *with order at least two if started sufficiently close to* $\mathbf{x}^*$*.* □

EXAMPLE 2.8. Solve the following problem by Newton's method

$$\text{Minimize} \quad f(\mathbf{x}) = (x_1 - 2)^4 + (x_1 - 2x_2)^2.$$

The termination scalar is 0.045 and the starting point is $(0, 3)^T$.

SOLUTION. At each iteration, $\mathbf{x}_{k+1}$ is given by $\mathbf{x}_{k+1} = \mathbf{x}_k - [\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$. After six iterations, the point $\mathbf{x}_6 = (1.83, 0.91)^T$ is reached. The algorithm is terminated since $\|\nabla f(\mathbf{x}_6)\| = 0.04 < 0.045$. The points generated by the method are shown in the following table.

Iteration k	$\mathbf{x}_k$ $f(\mathbf{x}_k)$	$\nabla f(\mathbf{x}_k)$	$\nabla^2 f(\mathbf{x}_k)$	$[\nabla^2 f(\mathbf{x}_k)]^{-1}$	$-[\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$	$\mathbf{x}_{k+1}$
0	(0.00, 3.00) 52.00	(−44.0, 24.0)	$\begin{bmatrix} 50.0 & -4.0 \\ -4.0 & 8.0 \end{bmatrix}$	$\frac{1}{384}\begin{bmatrix} 8.0 & 4.0 \\ 4.0 & 50.0 \end{bmatrix}$	(0.67, −2.67)	(0.67, 0.33)
1	(0.67, 0.33) 3.13	(−9.39, −0.04)	$\begin{bmatrix} 23.23 & -4.0 \\ -4.0 & 8.0 \end{bmatrix}$	$\frac{1}{169.84}\begin{bmatrix} 8.0 & 4.0 \\ 4.0 & 23.23 \end{bmatrix}$	(0.44, 0.23)	(1.11, 0.56)
2	(1.11, 0.56) 0.63	(−2.84, −0.04)	$\begin{bmatrix} 11.50 & -4.0 \\ -4.0 & 8.0 \end{bmatrix}$	$\frac{1}{76}\begin{bmatrix} 8.0 & 4.0 \\ 4.0 & 11.50 \end{bmatrix}$	(0.30, 0.14)	(1.41, 0.70)
3	(1.41, 0.70) 0.12	(−0.80, −0.04)	$\begin{bmatrix} 6.18 & 4.0 \\ -4.0 & 8.0 \end{bmatrix}$	$\frac{1}{33.44}\begin{bmatrix} 8.0 & 4.0 \\ 4.0 & 6.18 \end{bmatrix}$	(0.20, 0.10)	(1.61, 0.80)
4	(1.61, 0.80) 0.02	(−0.22, −0.04)	$\begin{bmatrix} 3.83 & -4.0 \\ -4.0 & 8.0 \end{bmatrix}$	$\frac{1}{14.64}\begin{bmatrix} 8.0 & 4.0 \\ 4.0 & 3.83 \end{bmatrix}$	(0.13, 0.07)	(1.74, 0.87)
5	(1.74, 0.87) 0.005	(−0.07, 0.00)	$\begin{bmatrix} 2.81 & -4.0 \\ -4.0 & 8.0 \end{bmatrix}$	$\frac{1}{6.48}\begin{bmatrix} 8.0 & 4.0 \\ 4.0 & 2.81 \end{bmatrix}$	(0.09, 0.04)	(1.83, 0.91)
6	(1.83, 0.91) 0.0009	(0.0003, −0.04)				

<u>Drawbacks</u>

Steepest Descent — Zigzagging phenomenon.

Newton's Method — High computational complexity.